UNIVERSITY OF MORATUWA

Faculty of Engineering

Department of Electronic and Telecommunication Engineering

B.Sc. Engineering

Semester 8 Examination

EN4730 CONVEX ENGINEERING DESIGN

Time allowed: **Two (2)** hours June 2025

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 5 pages.
- This examination accounts for **30%** of the module assessment. The total maximum mark attainable is **100**. The marks assigned for each question & sections there of are indicated in brackets.
- This is a closed-book examination.
- Electronic/communication devices are not permitted. Only equipment allowed is a calculator approved and labeled by the Faculty of Engineering.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.
- Keep your answers **brief** and to the point.

ADDITIONAL MATERIAL:

- No additional materials are provided.

Continued ...

Question 1.

(a) Determine whether the given functions are convex, concave, or *neither convex nor concave*. If necessary, specify the *conditions on the parameters* under which convexity or concavity holds.

i. $\sum_{i=1}^{n} x_i \log_2\left(\frac{x_i}{t_i}\right)$ with constant $t_i > 0 \ \forall\, i$. [6 marks]

ii. $x_1 + \exp(x_1) - \exp(x_1 + x_2)$. [6 marks]

a) i) $f(x_i) = \sum_{i=1}^{n} x_i \log_2\left(\frac{x_i}{t_i}\right)$ $\qquad t_i > 0 \ \forall i$

$$f(x) = \sum_{i=1}^{n} x_i \left(\log_2 x_i - \log_2 t_i\right)$$

$$= \underbrace{\sum_{i=1}^{n} x_i \log_2 x_i}_{g(x_i)} - \underbrace{\sum_{i=1}^{n} x_i \log_2 t_i}_{h(x_i)}$$

$$x_i \log_2 x_i = \frac{x_i \ln x_i}{\ln 2}$$

$m(x) = x \ln x$

$m'(x) = 1 + \ln x$

$m''(x) = \frac{1}{x} > 0 \quad ; x > 0$

$m(x)$ convex

$\frac{m(x)}{\ln 2}$ convex

$g(x_i)$ convex for $x_i > 0$

$h(x_i) = \log_2 t_i \cdot x_i$

linear function of x_i

$h(x_i)$ - both convex and concave

$f(x_i)$ is convex $x_i > 0$, $t_i > 0$

ii) $f(x_1, x_2) = x_1 + \exp(x_1) - \exp(x_1 + x_2)$

$$\frac{\partial f}{\partial x_1} = 1 + e^{x_1} - e^{x_1 + x_2}$$

$$\frac{\partial^2 f}{\partial x_1^2} = e^{x_1} - e^{x_1 + x_2}$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1} = -e^{x_1 + x_2}$$

$$\frac{\partial f}{\partial x_2} = -e^{x_1 + x_2}$$

$$\frac{\partial^2 f}{\partial x_2^2} = -e^{x_1 + x_2}$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = -e^{x_1 + x_2}$$

$$H = \begin{bmatrix} e^{x_1} - e^{x_1 + x_2} & -e^{x_1 + x_2} \\ -e^{x_1 + x_2} & -e^{x_1 + x_2} \end{bmatrix}$$

$$|H| = (e^{x_1} - e^{x_1+x_2})(-e^{x_1+x_2}) - (e^{x_1+x_2})^2$$

$$|H| = -e^{2x_1+x_2} < 0 \qquad \text{indefinite}$$

neither convex nor concave

(b) For each of the sets (A, B and C) given in Figure Q1.1, determine whether each set is convex or not with justification. [3 marks]

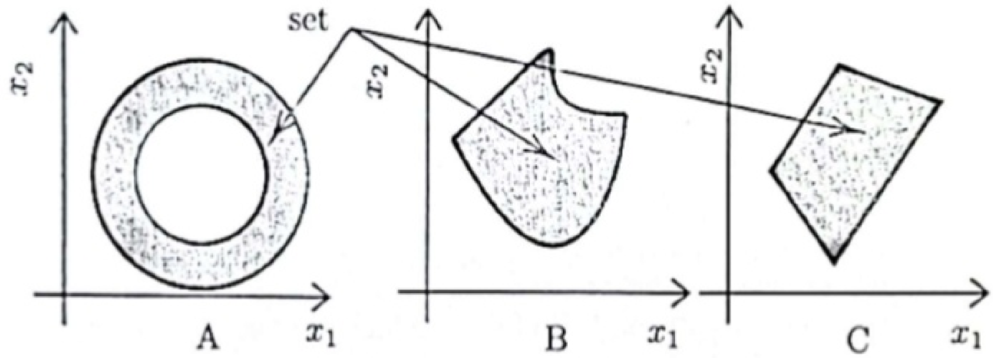
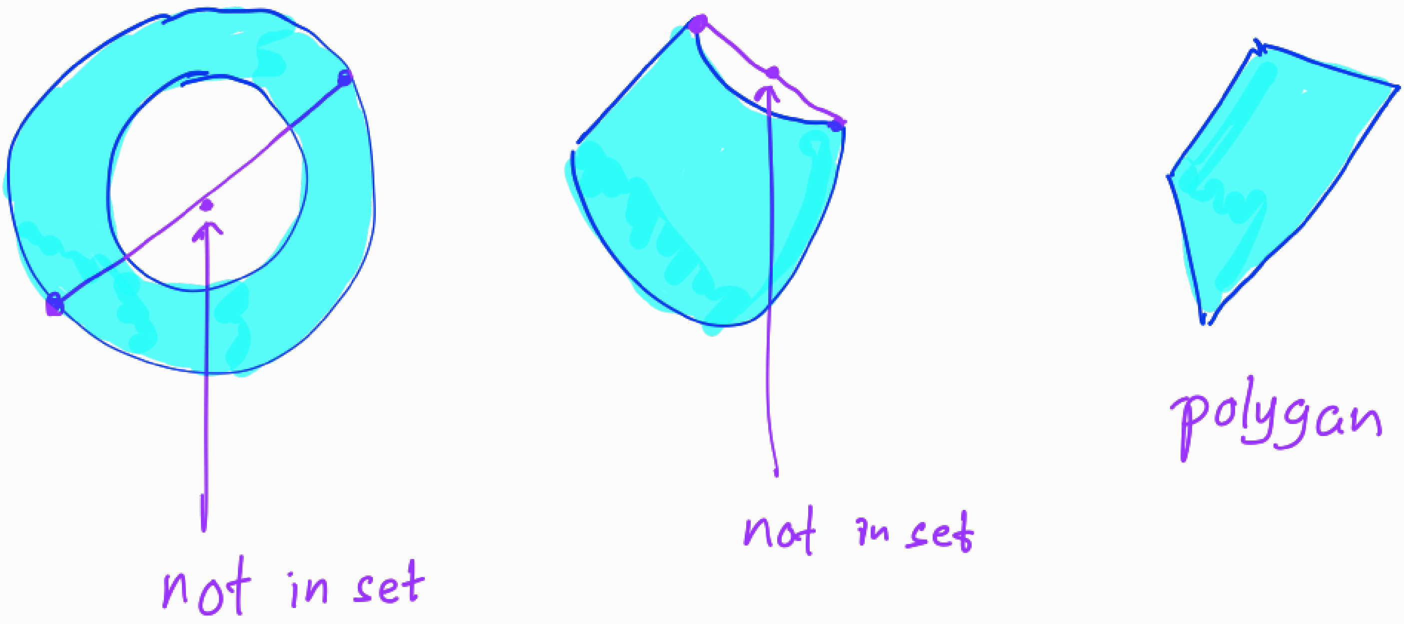


Figure Q1.1: Sets A, B and C.



non convex non convex convex

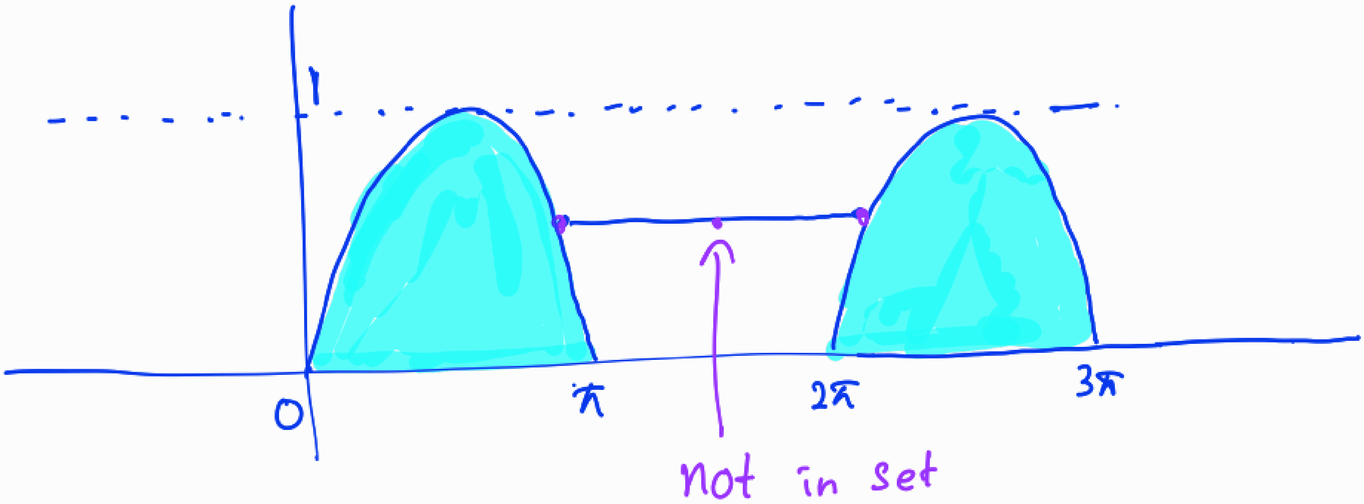
(c) Consider the following optimization problem with an arbitrary function $f_0(x)$

$$\begin{aligned} \text{minimize} \quad & f_0(x), \\ \text{subject to} \quad & 0 \leq \sin(x) \leq 1. \end{aligned}$$

i. Sketch the feasible set. [1 mark]

ii. Determine whether the feasible set is convex or not with justification. [2 marks]

iii. If the feasible set in Q1.(c) ii is not convex, what additional constraint(s) on x would make it convex ? [2 marks]



non convex

feasible set = $[0, \pi] \cap (2\pi, 3\pi] \cap (4\pi, 5\pi] \cap \ldots$

$= [2k\pi, (2k+1)\pi] \quad k \in \mathbb{Z}$

constraint = $x \in [0, \pi]$

limit domain to single interval

(d) Show that a function $f(x) \in C^2$ is convex over a convex set R_c if and only if, the Hessian $H(x)$ of $f(x)$ is positive semi-definite for all $x \in R_c$. Here, C^2 denotes the class of twice continuously differentiable functions.
Hint: $f(x+\delta) = f(x) + \nabla(x)^T\delta + \frac{1}{2}\delta^T H(x+\alpha\delta)\delta$ where $0 \le \alpha \le 1$ and δ is a small perturbation vector. [5 marks]

$$f(x+\delta) = f(x) + \nabla f(x)^T \delta + \frac{1}{2}\delta^T H(x+\alpha\delta)\delta$$

$H(x)$ PSD $\Rightarrow$ $f(x)$ convex

Take $x_1 = x + d$,

$$f(x_1) = f(x) + \nabla f(x)^T (x_1 - x) + \frac{1}{2} d^T H(x + d\alpha) d$$

Since $H(x)$ is PSD,

$$\frac{1}{2} d^T H(x+d\alpha) d \geq 0 \qquad \boxed{v^T H v \geq 0}$$

$$f(x_1) \geq f(x) + \nabla f(x)^T (x_1 - x)$$

This is first order condition of convexity.

$\therefore$ $f(x)$ is convex.

$f(x)$ convex $\Rightarrow$ $H(x)$ PSD

$H(x)$ not PSD $\Rightarrow$ $f(x)$ not convex

$$\boxed{a \Rightarrow b \;=\; \sim b \Rightarrow \sim a}$$

Since $H(x)$ is not PSD,

$$\frac{1}{2} d^T H(x+d\alpha) d < 0$$

$$f(x_1) < f(x) + \nabla f(x)^T (x_1 - x)$$

$\therefore$ $f(x)$ is not convex

Original statement is true.

$H(x)$ PSD $\Leftrightarrow$ $f(x)$ convex

Question 2

(a) Consider the following linear programming (LP) problem:

$$\begin{aligned} \text{maximize} \quad & 3x_1 + 2x_2, \\ \text{subject to} \quad & 4x_1 + 2x_2 \le 18, \\ & x_1 + 2x_2 \le 8, \\ & x_1 + x_2 \le 5, \\ & x_1, x_2 \ge 0. \end{aligned}$$

i. What is the dual problem of the given linear program (LP)? **[2 marks]**

ii. It is given that for the primal problem, the complementary slackness variables are $s_1 = 0$, $s_2 = 2$, and $s_3 = 0$. These slack variables are used to convert the inequality constraints into equality constraints. Show that $x_1^* = 4$, $x_2^* = 1$ is an optimal solution. **[10 marks]**

iii. Show that strong duality holds. **[5 marks]**

Hint: For complementary slackness, $\lambda_i s_i = 0 \quad \forall i$, and $x_j z_j = 0 \quad \forall j$. Here, s_i and z_j are slack variables. The primal and dual variables are x_j and λ_i, respectively.

max $\quad c^T x$

$$\begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad c = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

subject to $\quad Ax \le b$

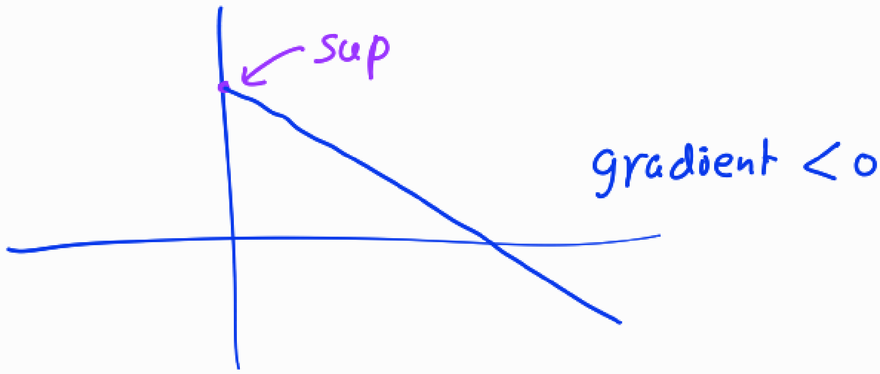
$$\begin{bmatrix} 4 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \le \begin{bmatrix} 18 \\ 8 \\ 5 \end{bmatrix}$$

$$L(x, \lambda) = c^T x + \lambda^T (b - Ax)$$

$$V(x, \lambda) = \sup_{x \geq 0} \{ L(x, \lambda) \}$$

$$= \sup_{x \geq 0} \{ c^T x + \lambda^T (b - Ax) \}$$

$$= \sup_{x \geq 0} \{ \lambda^T b + (c^T - \lambda^T A) x \}$$



$$g(\lambda) = \begin{cases} \lambda^T b, & \lambda^T A \geq c^T \\ \infty, & \text{otherwise} \end{cases}$$

$$\min \quad \lambda^T b$$

$$\text{subject to} \quad \lambda^T A \geq c^T$$
$$\lambda \geq 0$$

$$\lambda^T b = [\lambda_1 \ \lambda_2 \ \lambda_3] \begin{bmatrix} 18 \\ 8 \\ 5 \end{bmatrix}$$

$$\lambda^T b = 18\lambda_1 + 8\lambda_2 + 5\lambda_3$$

$$\lambda^T A \geq c^T$$

$$\begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 \end{bmatrix} \begin{pmatrix} 4 & 2 \\ 1 & 2 \\ 1 & 1 \end{pmatrix} \geq \begin{bmatrix} 3 & 2 \end{bmatrix}$$

$$4\lambda_1 + \lambda_2 + \lambda_3 \geq 3$$

$$2\lambda_1 + 2\lambda_2 + \lambda_3 \geq 2$$

min $\quad 18\lambda_1 + 8\lambda_2 + 5\lambda_3$

subject to $\quad 4\lambda_1 + \lambda_2 + \lambda_3 \geq 3$

$\quad 2\lambda_1 + 2\lambda_2 + \lambda_3 \geq 2$

$\quad \lambda_1, \lambda_2, \lambda_3 \geq 0$

primal

$$4x_1 + 2x_2 \leq 18$$

$$x_1 + 2x_2 \leq 8$$

$$x_1 + x_2 \leq 5$$

$$4x_1 + 2x_2 + S_1 = 18$$

$$x_1 + 2x_2 + S_2 = 8$$

$$x_1 + x_2 + S_3 = 5$$

$$4x_1 + 2x_2 = 18 \quad - ①$$

$$x_1 + 2x_2 + 2 = 8$$

$$x_1 + 2x_2 = 6 \quad - ②$$

$$x_1 + x_2 = 5 \quad - ③$$

$$x_1^* = 4 \;, \quad x_2^* = 1$$

dual

$$4\lambda_1 + \lambda_2 + \lambda_3 \geq 3$$

$$2\lambda_1 + 2\lambda_2 + \lambda_3 \geq 2$$

$$4\lambda_1 + \lambda_2 + \lambda_3 + z_1 = 3$$

$$2\lambda_1 + 2\lambda_2 + \lambda_3 + z_2 = 2$$

Complementary slackness

$$\lambda_i S_i = 0$$

$$\lambda_1 S_1 = 0 \qquad \lambda_2 S_2 = 0 \qquad \lambda_3 S_3 = 0$$

$$\lambda_1 (0) = 0 \qquad \lambda_2 (2) = 0 \qquad \lambda_3 (0) = 0$$

$$\lambda_2 = 0$$

$$x_j z_j = 0$$

$$x_1 z_1 = 0 \qquad\qquad x_2 z_2 = 0$$

$$(4) z_1 = 0 \qquad\qquad (1) z_2 = 0$$

$$z_1 = 0 \qquad\qquad z_2 = 0$$

$$4\lambda_1 + \lambda_3 = 3 \quad — (4)$$

$$2\lambda_1 + \lambda_3 = 2 \quad — (5)$$

$$\lambda_1^* = 0.5 \qquad \lambda_3^* = 1$$

Finding objective

primal $\qquad 3x_1 + 2x_2$

$$p^* = 3(4) + 2(1) = 14$$

dual $\qquad 18\lambda_1 + 8\lambda_2 + 5\lambda_3$

$$d^* = 18(0.5) + 8(0) + 5(1) = 9 + 5 = 14$$

$$p^* = d^* = 14$$

strong duality holds.

(b) Consider the following optimization problem.

$$\underset{x}{\text{minimize}} \quad [-4 \ -2]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

$$\text{subject to} \quad \begin{bmatrix} 1 & 1 \\ 30 & 10 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 50 \\ 700 \\ 56 \\ 0 \\ 0 \end{bmatrix}.$$

i. Identify the vertices of the feasible region and determine which of them could be an optimal solution. [3 marks]

ii. Using geometric arguments, show that there are infinite number of solutions for this problem. [5 marks]

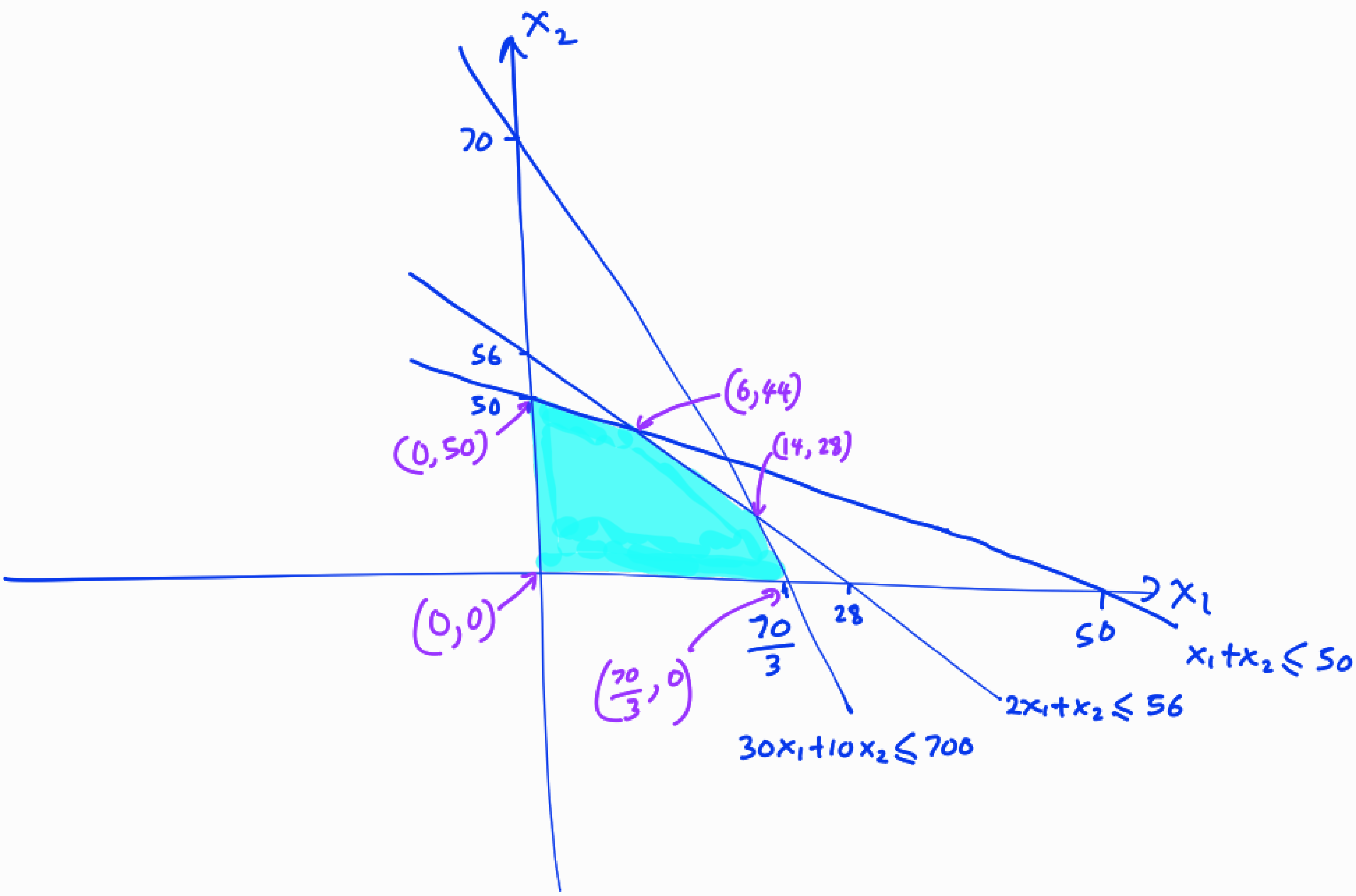
$x_1 + x_2 \leqslant 50$

$30x_1 + 10x_2 \leqslant 700$

$2x_1 + x_2 \leqslant 56$

$-x_1 \leqslant 0$

$-x_2 \leqslant 0$



i) $(0,0)$, $(0,50)$, $\left(\frac{70}{3}, 0\right)$, $(6,44)$, $(14,28)$

$f(x) = -4x_1 - 2x_2$

$(0,0) \rightarrow 0$

$(0,50) \rightarrow -100$

$\left(\frac{70}{3}, 0\right) \rightarrow -93.3$

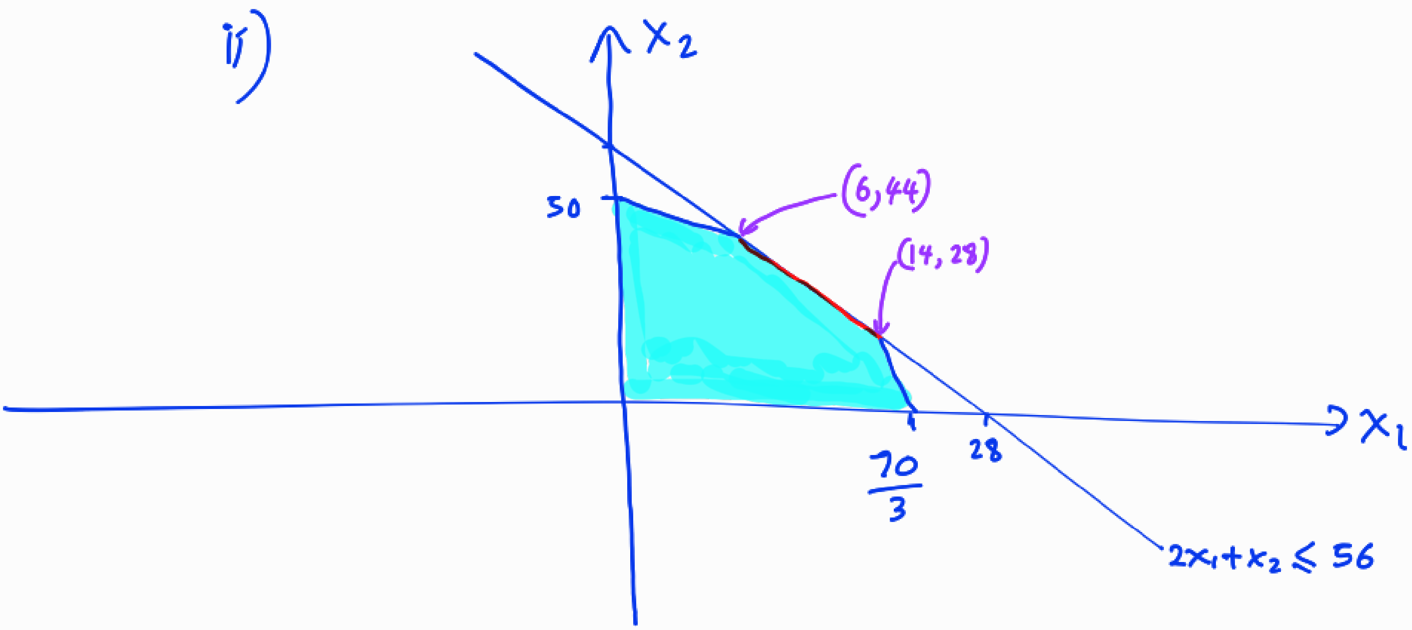
$(6,44) \rightarrow -112$

$(14,28) \rightarrow -112$

min value $= -112$

optimal vertices $= (6,44)$, $(14,28)$

ii)



$f(x) = -4x_1 - 2x_2$

objective function is parallel to $2x_1 + x_2 = 56$

$f(6,44) = -4(6) - 2(44) = -112$

$f(14,28) = -4(14) - 2(28) = -112$

$f(10,36) = -4(10) - 2(36) = -112$

all points between $(6,44)$ and $(14,28)$ on $2x_1 + x_2 = 56$ line gives same optimal value.

$\therefore$ infinitely many solutions exist.

$$\begin{aligned}
\underset{\boldsymbol{x}}{\text{minimize}} \quad & f(\boldsymbol{x}) \\
\text{subject to} \quad & h_i(\boldsymbol{x}) \le 0, \quad i = 1, \dots, m \\
& g_i(\boldsymbol{x}) = 0, \quad i = 1, \dots, n
\end{aligned}$$

Let $\boldsymbol{x}^\star$ and $(\boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)$ be primal and dual optimal points with zero duality gap. Since $\boldsymbol{x}^\star$ minimizes $L(\boldsymbol{x}, \boldsymbol{\lambda}^\star, \boldsymbol{\mu}^\star)$ over $\boldsymbol{x}$, its gradient vanishes at $\boldsymbol{x}^\star$:

$$\nabla f(\boldsymbol{x}^\star) + \sum_{i=1}^{n} \lambda_i^\star \nabla g_i(\boldsymbol{x}^\star) + \sum_{i=1}^{m} \mu_i^\star \nabla h_i(\boldsymbol{x}^\star) = 0.$$

Thus, the Karush-Kuhn-Tucker (KKT) conditions are:

$$\begin{cases}
h_i(\boldsymbol{x}^\star) \le 0, & i = 1, \dots, m, \text{(Feasibility)} \\
g_i(\boldsymbol{x}^\star) = 0, & i = 1, \dots, n, \text{(Feasibility)} \\
\mu_i^\star \ge 0, & i = 1, \dots, m, \text{(Dual feasibility)} \\
\mu_i^\star h_i(\boldsymbol{x}^\star) = 0, & i = 1, \dots, m, \text{ (Complementary slackness)} \\
\nabla f(\boldsymbol{x}^\star) + \sum_{i=1}^{n} \lambda_i^\star \nabla g_i(\boldsymbol{x}^\star) + \sum_{i=1}^{m} \mu_i^\star \nabla h_i(\boldsymbol{x}^\star) = 0, & \text{(Stationary).}
\end{cases}$$

Question 3.

Consider a portfolio optimization problem where the goal is to maximize the risk-adjusted return. This problem is formulated as:

$$\begin{aligned}
\underset{w}{\text{maximize}} \quad & \mu^T w - \gamma w^T \Sigma w, \\
\text{subject to} \quad & 1^T w = 1, \\
& w_i > 0, \quad i = 1, \ldots, n, \\
& \gamma > 0.
\end{aligned}$$

Here, $w \in \mathbb{R}^n$ is the portfolio fractional allocation vector, where each element represents the proportion of the total investment allocated to asset i, $\mathbf{\Sigma}$ is the covariance matrix of asset returns and γ is the risk aversion parameter, respectively. An investor wants to allocate funds between two stocks with expected returns of $\mu_1 = 8\%$ and $\mu_2 = 6\%$. The covariance matrix for their returns is $\Sigma = \begin{bmatrix} 0.04 & 0.01 \\ 0.01 & 0.09 \end{bmatrix}$, and $\gamma = 1$.

(a) Write down the Lagrangian function and Karush–Kuhn–Tucker (KKT) conditions. [5 marks]

(b) Use the Karush–Kuhn–Tucker (KKT) conditions to find the optimal portfolio fractional allocation (w_1, w_2) that maximizes the risk-adjusted return (For the sake of simplicity, you may ignore the constraints of w_1 and w_2 being non-negative). [15 marks]

(c) Formulate an optimization problem for the worst-case risk analysis. [5 marks]

a) $L(w, \lambda) = \mu^T w - \gamma w^T \Sigma w + \lambda(1^T w - 1)$

stationarity $\quad \nabla_w L = 0 \quad — \ 1$

primal feasibility $\quad w_1 + w_2 = 1 \quad — \ 2$

complementary slackness $\quad$ no inequality constraints

dual feasibility $\quad \mu_i w_i = 0$

b) $$L(w,\lambda) = \mu^T w - \gamma w^T \Sigma w + \lambda(1^T w - 1)$$

$$\nabla_w L = \mu - \gamma \cdot 2\Sigma w + \lambda$$

$$0 = \mu - 2\Sigma w + \lambda$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} - 2\begin{bmatrix} 0.04 & 0.01 \\ 0.01 & 0.09 \end{bmatrix}\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} \lambda \\ \lambda \end{bmatrix}$$

$$0 = 0.08 - 0.08 w_1 - 0.02 w_2 + \lambda \quad — ③$$

$$0 = 0.06 - 0.02 w_1 - 0.18 w_2 + \lambda \quad — ④$$

③ - ④ $$0 = -0.02 + 0.06 w_1 - 0.16 w_2$$

② $\Rightarrow$ $$0 = -0.02 + 0.06 w_1 - 0.16(1 - w_1)$$

$$0 = -0.18 + 0.22 w_1$$

$$w_1 = \frac{9}{11} \qquad w_2 = \frac{2}{11}$$

c) minimize worst case risk

$$\min_{w} \max_{\Sigma \in S} (w^T \Sigma w)$$

s.t. $$\Sigma \geq 0$$
$$1^T w = 1$$
$$w \geq 0 \qquad i = 1, 2, \ldots, n$$

Question 4.

(a) Consider the following Quadratic Programming (QP) problem:

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & \frac{1}{2}x^T P x + q^T x, \\ \text{subject to} \quad & Ax - b = 0, \end{aligned}$$

where x, $q, \in \mathbb{R}^n$, $b \in \mathbb{R}^m$ $A \in \mathbb{R}^{m\times n}$ and $P \in \mathbb{R}^{n\times n}$.
Show that

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix}_{(m+n)\times(m+n)} \begin{bmatrix} x \\ \lambda \end{bmatrix}_{(m+n)\times 1} = \begin{bmatrix} -q \\ b \end{bmatrix}_{(m+n)\times 1}$$

Here, $\lambda \in \mathbb{R}^m$ are Lagrangian multipliers. [10 marks]

$$\min \quad \frac{1}{2} x^T P x + q^T x$$

$$\text{subject to} \quad Ax - b = 0$$

$$L(x, \lambda) = \frac{1}{2} x^T P x + q^T x + \lambda^T (Ax - b)$$

$$L(x, \lambda) = \frac{1}{2} x^T P x + (q + A^T \lambda)^T x - \lambda b^T$$

$$\nabla_x L(x, \lambda) = Px + q + A^T \lambda$$

$$0 = Px + q + A^T \lambda$$

$$Px + A^T \lambda = -q \quad — (1)$$

$$Ax = b$$

$$Ax + 0\lambda = b \quad — (2)$$

$$\begin{pmatrix} P & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} -q \\ b \end{pmatrix}$$

(b) Solve the following Quadratic Programming (QP) problem using *the active set method* and determine the optimal x_1 and x_2. [15 marks]

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & 2x_1^2 + x_2^2 + 2x_2, \\ \text{subject to} \quad & g_1(x) = x_1 + x_2 \geq 5, \\ & g_2(x) = x_1 + x_2 \leq 8. \end{aligned}$$

	optimal point	function value	feasibility
Both constraints active	—	—	infeasible
Only g_1 active	$(2,3)$	23	feasible
Only g_2 active	$(3,5)$	53	feasible
Both constraints inactive	$(0,-1)$	-1	infeasible

Only g_1 active

$x_1 + x_2 = 5$

$f(x_1, x_2) = 2x_1^2 + x_2^2 + 2x_2$

$f(x_2) = 2(5 - x_2)^2 + x_2^2 + 2x_2$

$\nabla f(x_2) = -4(5 - x_2) + 2x_2 + 2$

$0 = -18 + 6x_2$

$x_2 = 3, \quad x_1 = 2$

$f(2,3) = 2(2)^2 + 3^2 + 2(3) = 23$

$2 + 3 \geq 5 \qquad 2 + 3 \leq 8$

$(2,3)$ is feasible

Only g_2 active

$x_1 + x_2 = 8$

$f(x_1, x_2) = 2x_1^2 + x_2^2 + 2x_2$

$f(x_2) = 2(8 - x_2)^2 + x_2^2 + 2x_2$

$\nabla f(x_2) = -4(8 - x_2) + 2x_2 + 2$

$0 = -30 + 6x_2$

$x_2 = 5, \quad x_1 = 3$

$f(3,5) = 2(3)^2 + 5^2 + 2(5) = 53$

$3 + 5 \geq 5 \qquad 3 + 5 \leq 8$

$(3,5)$ is feasible

Both constraints inactive

$$f(x_1, x_2) = 2x_1^2 + x_2^2 + 2x_2$$

$$\frac{\partial f}{\partial x_1} = 4x_1 \qquad\qquad \frac{\partial f}{\partial x_2} = 2x_2 + 2$$

$$0 = 4x_1 \qquad\qquad 0 = 2x_2 + 2$$

$$x_1 = 0 \qquad\qquad x_2 = -1$$

$$f(0,-1) = 2(0)^2 + (-1)^2 + 2(-1) = -1$$

$$0 + (-1) \geq 5$$

$$-1 \geq 5$$

$(0, -1)$ is infeasible